Creepypasta Narrative Analysis Using Python, Machine Learning, RapidMiner, and MongoDB

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Project Name - Creepypasta Narrative Analysis Using Python, Machine Learning, RapidMiner, and MongoDB

Project explanation:

This project analyzes an extensive set of creepypasta stories from the perspective of several layers of analysis that are aimed at uncovering the structural regularities inherent to the narrative data set. Despite being a set of stories, the data set includes elements of numerical, categorical, and temporal nature, which gives it a systemic nature. This analysis will hone in on how different dimensions relate to one another (particularly the numerical categories). Essentially, the structure of the dataset will be explored alongside the content.

This can be achieved through the integration of several analytics platforms: Python for EDA and models, RapidMiner for visual machine learning pipelines, and MongoDB to perform query operations at the document level. Each method provides an individual point of view: Python reveals the numerical topography of the data set, RapidMiner provides validation of the model-building process through its visual capabilities, and MongoDB recaptures the narrative aspect of the story by treating it as a document located in a thematic web. This work is thus not just the application of a single method but a holistic combination of methods that will shed light on the structure of the data set as a whole.

Conclusions:

EDA and Visualization

Exploratory data analysis showed that the creepypasta dataset has an architectural basis on the time dimension. Overall, the average score of creepypastas was found to be 7.56, while the average time to read a story amounted to 9.65 minutes; the correlation between reading time and story length turned out to be quite high (0.71). Visualizing this architecture, histograms indicated a large number of stories with 5-15 minutes reading time, whereas scatterplots showed the presence of a moderate correlation (0.34) between story length and its rating. The correlation matrix clearly illustrated the presence of such structural dimensions, implying that narrative size is one of those.

Machine Learning

The process of applying machine learning helped identify the nature of numerical features in terms of their behavior in model development. The k-means algorithm helped differentiate the documents based on their duration in three temporal classes, namely short (less than 5 minutes), medium (from 5 to 20 minutes), and long (more than 20 minutes), just like the distribution observed in the EDA phase. In turn, KNN classification demonstrated good results when classifying stories based on extremes, such as Distorted Warning Signals (3 minutes, 9.12 rating), but had some problems with mid-level ratings due to qualitative differences in themes.

RapidMiner

RapidMiner was used as an alternative analytical tool that verified the findings from Python via its graphical workflow. The reimplementation of the K-means model, KNN classifier, and performance analysis showed that the tendencies observed in the Python notebook persisted: temporal clustering, challenges predicting middle-range scores, and numerical consistency of the data set. Through its block-based approach, RapidMiner demonstrated that the analytical process can be easily transported to other platforms without being bound to one specific platform.

MongoDB

The narrative dimension was reintroduced through MongoDB by looking at individual stories as documents rather than rows of tables. Aggregation pipelines helped determine the thematic distribution in the corpus, where some themes occur more often than others, such as Beings and Entities (almost 300 occurrences) and Strange and Unexplained (over 250 occurrences). Regex search identified the motifs, which were constantly occurring in the corpus, like night and ghost. Structural outliers could be easily spotted, too, such as very long stories, such as 'Tales From an Ex-Convict' (201 minutes), and short stories with high ratings.

Final Conclusions

As a whole, these methods of analysis combine into one analytical framework, where each method helps to examine a different side of this dataset: Python sheds light on its numeric anatomy, machine learning helps us to see how this anatomy is structured, RapidMiner verifies whether the observed patterns are reliable, and finally, MongoDB allows restoring the semantic depth, which numeric models are not able to reflect. This study proves that the set of creepypastas works as a multi-layered system, whose complexity can be understood only from different perspectives.